Hierarchical Physical-Chain Decoupling With Geo-Semantic MoLoRA for All-in-One Multimodal Remote Sensing Image Restoration

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***Abstract*—All-in-one remote sensing (RS) multimodal image restoration has emerged as a promising paradigm for addressing diverse degradations within a unified framework. However, existing methodologies encompassing prompt-based, agent-based, and low-rank adapters (LoRA)-based approaches encounter a subtle architectural tradeoff. They often prioritize task heterogeneity at the expense of underlying physical homologies, or conversely, focus on cross-task shared priors while potentially overlooking unique task-specific heterogeneities. Furthermore, these systems can involve considerable computational overhead when addressing mixed degradations due to the sequential invocation of multiple tools. To overcome these limitations, we propose an approach named hierarchical physical-chain decoupling (HPCD), aiming to simultaneously exploit physical homology and task heterogeneity. A novel framework decomposes restoration into three progressive stages according to physical priors: global geographic layout learning in Stage I, physics-driven group homology extraction across atmospheric, sensor, and transmission chains in Stage II, and task-specific atomic expert refinement in Stage III. Specifically, we implement the refinement stage via mixture-of-experts low-rank adaptation (MoLoRAs), which enables dynamic expert activation based on identified degradation types. We further introduce a vision–language model (VLM)-guided regularization paradigm to mitigate diffusion-based hallucinations and develop Multi-LoRA distribution matching distillation (Multi-LoDMD) for efficient one-step inference. Finally, we establish UniRS-60K, a comprehensive benchmark for multimodal and adversarial restoration. Experiments demonstrate that HPCD achieves state-of-the-art (SOTA) precision and task compatibility across three benchmarks. The code and UniRS-60K dataset are released at https://github.com/zhentao-zou/HPCD**

***Index Terms*—All-in-one restoration, generative diffusion models, physical-chain decoupling, remote sensing (RS) images.**

Received 6 February 2026; revised 29 April 2026; accepted 5 May 2026. Date of publication 13 May 2026; date of current version 27 May 2026. This work was supported by the National Natural Science Foundation of China under Grant 62571315. *(Corresponding authors: Xiaogang Yu; Xue Jiang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3692975

I. Introduction

With the rapid advancement of satellite remote sensing (RS) technology, the proliferation of high-resolution satellite data has played an indispensable role in global environmental monitoring and change research. However, RS images inevitably suffer from various forms of information degradation during acquisition and transmission, stemming from atmospheric scattering, sensor limitations, and digital processing artifacts [1], [2]. Specifically, atmospheric scattering manifests as cloud cover and fog, sensor-related issues result in stripes and missing data requiring inpainting, and digital transmission introduces Gaussian noise, blur, and JPEG compression artifacts. These complex degradations often occur in isolation or coexist as composite distortions, significantly deteriorating the visual quality of RS imagery. Such degradation further constrains the performance of critical downstream applications, including change detection [3], ground object detection [4], and land-cover classification [5], [6]. Despite these challenges, a unified framework capable of restoring RS images across these diverse degradation types has not yet been explored. Consequently, it is imperative to develop a robust and effective framework to process these seven individual degradations and their complex mixed compositions, thereby facilitating high-quality RS scene restoration and analysis.

Traditional RS image restoration approaches have primarily relied on filter-based techniques and optimization-based variational methods [7], [8]. While these methods can achieve reasonable restoration performance under specific conditions, they are often constrained by rigorous mathematical assumptions. Consequently, they frequently yield unsatisfactory results when applied to complex, real-world scenarios. This limitation stems from the inherent inability of simplified mathematical models to adequately capture the sophisticated structural and spectral characteristics of diverse RS imagery, as illustrated in Fig. 2. In contrast, the rapid evolution of deep learning (DL) has ushered in a new paradigm of data-driven restoration. Various architectures, including convolution-based networks [9], transformer-based models [10], and more recently, diffusion-based frameworks [11], [12], have been proposed to reconstruct high-quality RS images from their corrupted counterparts. These methods leverage

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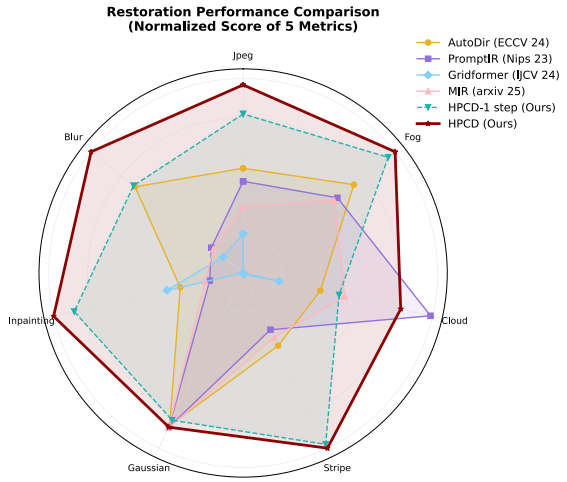


Fig. 1. Radar chart illustrating the normalized performance profiles of the proposed methods against SOTA all-in-one restoration baselines. Our framework demonstrates robust generalization across diverse degradation scenarios. Notably, the one-step distilled variant maintains high performance consistency with the full teacher model.

deep hierarchical features to adaptively model complex degradations that traditional variational approaches fail to resolve.

Recently, the research community has witnessed a paradigm shift toward all-in-one image restoration [13], which seeks to address multiple degradation types within a unified framework. Existing methodologies can be broadly categorized into agent-based systems [14], [15], prompt learning-based frameworks [16], [17], and mixture-of-experts (MoEs) architectures [18]. Agent-based approaches intelligently perceive degradation types in degraded inputs to invoke specialized pretrained networks; however, they often encounter challenges in fully exploiting the latent synergies between diverse degradations. Moreover, as these systems typically scale in complexity with the task count, optimizing parameter efficiency and inference latency remains a significant consideration for composite degradations. Similarly, prompt-based methodologies have advanced multitask restoration by integrating task-specific tokens into shared backbones. However, the intricate nature of RS scenarios can lead to representational competition within the shared latent space. Such tension between divergent objectives, for example, noise suppression versus texture synthesis, may occasionally restrict the model from reaching expert-level precision across all tasks, a phenomenon often characterized as negative transfer. In parallel, direct-low-rank adapters (LoRA)-based approaches utilize independent adapters to manage specific degradations. A potential opportunity for further development lies in more effectively leveraging the physical homologies shared across different tasks. Without ensuring distributional alignment across decoupled modules, achieving reliable feature generalization remains difficult, especially within the constraints of data-scarce environments or unforeseen out-of-distribution settings. In reality, certain RS tasks are intrinsically linked through shared physics. For instance, learning to defog exerts a positive impact on cloud removal, as both processes involve reversing atmospheric scattering to restore surface reflectance [7], [19], as illustrated in Fig. 2. Despite these substantial contributions, establishing a unified framework that seamlessly harmonizes physical homology and task heterogeneity remains an essential yet underexplored frontier in all-in-one multimodal RS image restoration.

Physical Homology and Heterogeneity

Degradations in RS imagery, ranging from atmospheric interference to sensor limitations, exhibit fundamental physical homologies in the latent feature space while maintaining functional heterogeneity in their observation models.

Based on the observations above, we propose an HPCD method, which is designed to simultaneously exploit physical homology and task heterogeneity. As illustrated in Fig. 3, we divide the training process into three stages to effectively capture both physics-grouped shared features and task-specific details. In Stage I (global layout learning), we fine-tune the pretrained backbone on the entire corpus of degraded imagery to learn the global geographic layout and maintain semantic consistency. In Stage II (physics-driven grouped tuning), we decouple the LoRA modules into three distinct groups based on the RS imaging chain. The first group focuses on **Digital Transmission Degradations**, such as JPEG compression artifacts, Gaussian noise, and blur. From a signal-processing perspective, these share a commonality as they all contribute to the loss of high-frequency information and a reduction in signal fidelity during the acquisition and transmission pipeline [20]. The second group addresses **Sensor Acquisition Degradations**, where structured artifacts like stripe noise and data missing (inpainting) share the commonality of disrupting spatial–spectral continuity. Unlike random noise, these artifacts necessitate the capture of long-range dependencies and low-rank priors [21]. The third group is the **Atmospheric Scattering Group**, which encompasses cloud and haze removal. Specifically, haze and semitransparent thin clouds share identical physical characteristics in terms of transmission and air light estimation. Finally, in Stage III (task-specific expert refinement), we implement the refinement process via mixture-of-experts low-rank adaptation (MoLoRAs). Each group LoRA is partitioned into specialized atomic experts to learn fine-grained task details. It is important to note that, unlike conventional MoEs architectures that rely on learned routing gates lacking physical interpretability, our approach constructs a hierarchical taxonomy rooted in the physical RS imaging chain. By progressing from unified priors (Stage I) to groupwise physical homologies (Stage II) and ultimately to atomic expertise (Stage III), we systematically decouple intertask interference while maximizing cross-task synergy. This structured decoupling ensures high task compatibility and enables high-quality restoration results. The visualization results are shown in Fig. 4, and the comprehensive qualitative comparisons are presented in Fig. 1.

Owing to the inherent stochastic nature of diffusion models, the generation process may occasionally yield plausible geographic details that do not align with the original image semantics. To mitigate this, we introduce a vision–language model (VLM) to perceive specific degradations and predict geographic context priors. Unlike conventional degradation perception models, the integration of a VLM enables our framework to simultaneously identify corruptions and infer

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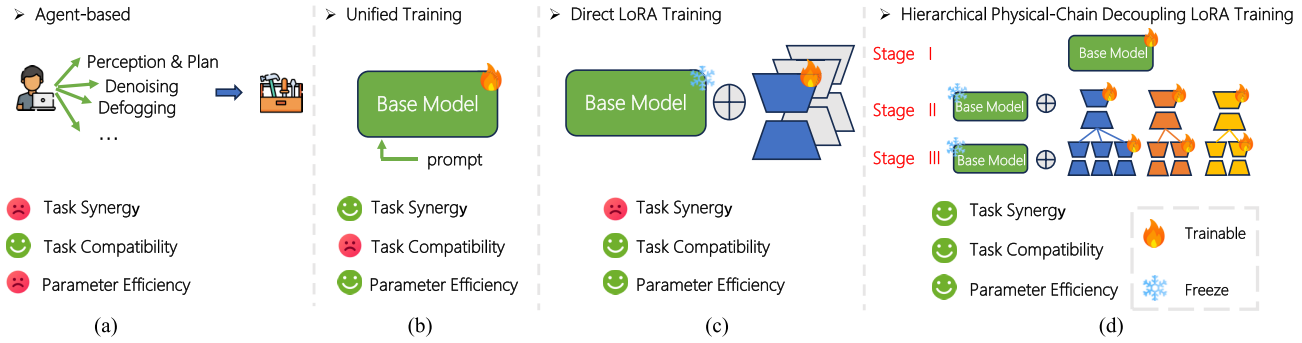


Fig. 2. Comparison of different training paradigms for all-in-one RS image restoration. (a) Agent-based approaches incur substantial inference overhead and lack robustness in mixed degradation scenarios due to the sequential invocation of multiple specialized networks. (b) Prompt-based and (c) direct LoRA training methods often suffer from a performance tradeoff between task synergy and task compatibility. (d) Our proposed HPCD framework simultaneously exploits physical homology and task heterogeneity. By decomposing the restoration process into three progressive stages based on physical priors, **HPCD** enables the model to capture physics-grouped shared features in stage II (e.g., understanding the atmospheric scattering model through joint cloud and fog training) while maintaining expert-level precision via task-specific refinement in stage III.

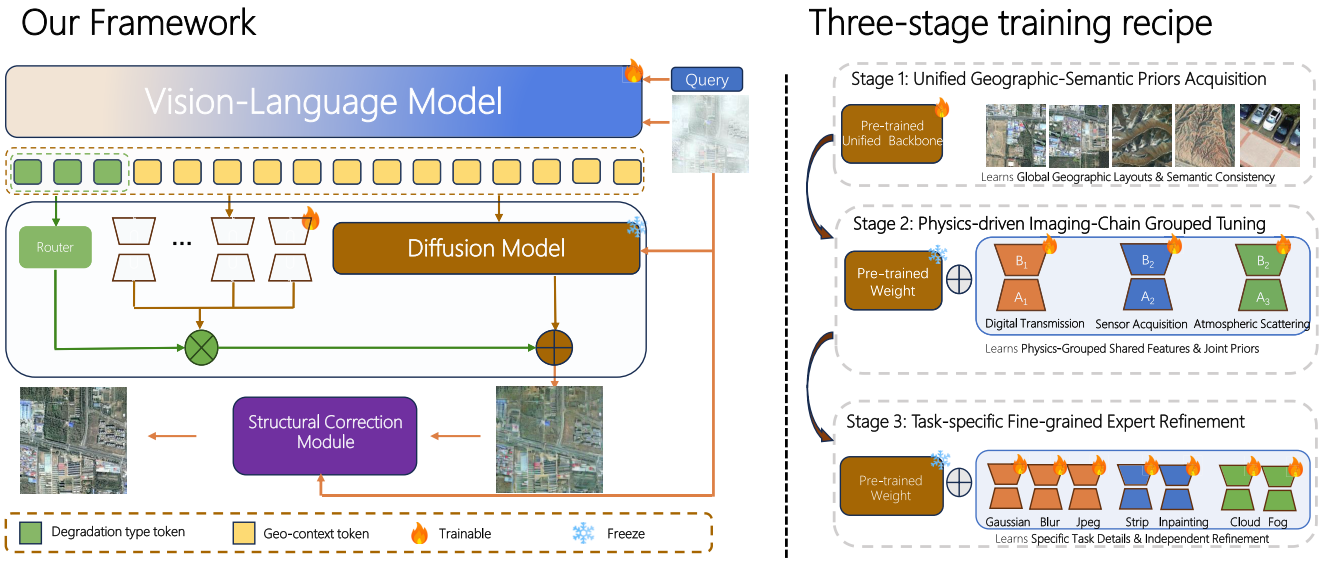


Fig. 3. Overview of the proposed **HPCD** framework and its three-stage training strategy. (a) Framework fine-tunes a VLM to extract a **degradation-type token** for task-specific expert activation and a **geo-context token** to provide semantic guidance for the diffusion process. (b) **Three-stage training recipe** progressively optimizes the network by exploiting physical homology in grouped degradations and addressing task heterogeneity through atomic expert refinement. In addition to task-specific restoration, we leverage the pretrained weights from Stage I to evaluate the model's capacity for **mixed degradation removal and adversarial defense.**

the underlying geographic context of the scene, thereby providing a robust foundational basis for conditional image generation. Furthermore, to accelerate inference speed and meet real-time requirements, we propose a novel Multi-LoRA distribution matching distillation (Multi-LoDMD) method specifically designed for diverse degradation scenarios. Overall, our architecture maintains low computational overhead while leveraging a single set of hierarchical parameters to capture both group-shared and task-specific features. This design effectively addresses diverse RS degradations, yielding high-fidelity restoration results. The main contributions of this work are summarized as follows.

1) *Hierarchical Physical-Chain Decoupling (HPCD) Framework:* We propose a three-stage restoration framework that aligns model training with the physical imaging chain. This architecture effectively exploits physical homologies across task groups while simultaneously addressing the fine-grained heterogeneity of individual degradations.
2) *VLM-Guided Semantic Regularization:* To mitigate hallucinations in diffusion-based generation, we introduce a regularization paradigm leveraging VLMs. By integrating geographic and degradation-aware priors, the framework ensures strict semantic fidelity and contextual consistency in the restored imagery.
3) *Efficient Multi-LoRA Distillation:* We develop Multi-LoDMD, a distribution-matching distillation strategy tailored for multitask scenarios. This enables rapid one-

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Fig. 4. Visual restoration results of the HPCD framework. Our method demonstrates robust performance across seven primary single degradations, various complex mixed distortions (e.g., Gaussian+fog and stripe+fog), and security-critical adversarial defense tasks.

step restoration across diverse degradations, significantly reducing computational latency without compromising high fidelity performance.

4) *The UniRS-60K Benchmark:* We present a comprehensive large-scale dataset featuring seven primary degradations and their composite forms. Distinctively, UniRS-60K incorporates multimodal data and adversarial scenarios, providing a rigorous foundation for evaluating robustness in security-critical applications.

II. Related Work

A. All-in-One Image Restoration

Image restoration methods have evolved from addressing specific degradations, such as denoising, dehazing, and deblurring, toward comprehensive all-in-one restoration. Early attempts at unified image restoration focused on parameterized image-operator frameworks, which laid the groundwork for modern universal architectures. With the advancement of DL, these methodologies can be broadly categorized into three types: prompt learning, LoRA-based adaptation, and agent-based systems.

Prompt-based methodologies [22], [23], [24] attempt to handle multitask restoration by injecting task-specific tokens into a shared backbone. Sun et al. [16] introduced AdaPrompt-IR, which combines a degradation representation codebook with prompt learning to implicitly guide the restoration of missing degradation semantics. Meanwhile, Zhang et al. [17] proposed Perceive-IR, a backbone-agnostic framework that utilizes CLIP-based prompt alignment to achieve fine-grained quality control across diverse degradation types and severity levels.

LoRA-based approaches [25] employ parameter-efficient fine-tuning to adapt pretrained models to multiple tasks. Zhang et al. [26] developed a universal framework based on multiple LoRAs that uses a similarity-based composing strategy to adaptively combine experts for mixed degradation restoration. Similarly, Ai et al. [18] proposed LoRA-IR, a flexible framework that leverages a MoEs architecture and a degradation-guided router to dynamically integrate multiple low-rank restoration experts. TCLRD [27] explicitly models the intrinsic low-rank and shift-invariant properties of the RS data itself. Although both TCLRD and LoRA-based approaches leverage low-rank properties, the fundamental distinction lies in the domain in which they operate. TCLRD focuses on the low rankness of the input RS data, whereas the LoRA framework exploits the low-rank nature of parameter updates (ΔW). This enables efficient model adaptation governed by structural weight constraints rather than explicit data-level priors, allowing the model to adapt to downstream tasks without necessitating specialized tensor-based reconstructions.

Agent-based systems [15], [28], [29] utilize autonomous workflows to orchestrate specialized tools. Jiang et al. [14] proposed MAIR, a multiagent approach that reverse-engineers real-world degradation priors through a collaborative team of scheduler and expert agents to reduce search space and inference costs. Furthermore, Chen et al. [15] introduced RestoreAgent, an intelligent system leveraging multimodal large language models to autonomously determine task sequences and select optimal models for restoring images with complex, multiple degradations. While current research primarily focuses on optical RS image restoration, extending the "all-in-one" paradigm to encompass broader multimodal contexts and 3-D spatial perception represents a compelling frontier [30], [31].

B. RS Image Restoration

The evolution of RS image restoration has progressed through three primary paradigms: traditional optimization-based approaches, DL-based methods, and the recent emergence of all-in-one frameworks.

Traditional approaches primarily rely on filters, low-rank models [32], and total-variation modules [33], [34] to exploit the physical characteristics of degradation [35], [36] and the inherent features of natural RS imagery. For instance, Li et al. [35] proposed a two-stage method combining homomorphic filtering and a sphere model-improved dark channel prior (DCP) to effectively remove uneven haze and thin clouds. Similarly, Li et al. [34] introduced graph-SSTV (GSSTV), a regularization model that utilizes a weighted spatial difference operator designed based on explicit graph structures to preserve complex spatial textures during mixed noise removal.

With the rapid advancement of DL, DL-based approaches [37], [38], [39], [40] have achieved impressive results by learning complex hierarchical features directly from large-scale data. Sun et al. [39] proposed the AIDTransformer, which utilizes a novel deformable multihead attention mechanism with a spatially attentive offset extractor to reconstruct fine-level textures and capture relevant contextual information for aerial image dehazing. Furthermore, Dong et al. [40] introduced a gradient-guided multiframe super-resolution (MFSR)

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framework that integrates learned gradient priors with nonlocal total variation to simultaneously enhance sharp edges and suppress visual artifacts.

Recently, a few researchers have explored all-in-one restoration in the RS scene. PhyDAE [41] introduces a physics-guided architecture that transforms implicit features into explicit decision signals to restore haze, noise, blur, and low-light conditions through physics-aware expert modules. Ada4DIR [42] proposes an adaptive model-driven network that integrates four distinct degradation models with a fusion transformer block, leveraging prompt learning to guide the restoration process. While recent studies such as PhyDAE [41] and Ada4DIR [42] have explored all-in-one RS image restoration, their scope remains limited to four specific degradation types, namely dehazing, deblurring, denoising, and low-light enhancement. However, real-world RS degradations extend significantly beyond these categories. To bridge this gap, our proposed HPCD framework decomposes restoration into three progressive stages, thus enabling the comprehensive handling of seven primary degradations along with their complex mixed compositions and security-critical adversarial defense scenarios.

III. Proposed Method

A. Vision–Language Prior Extraction

The accurate identification of degradation types **plays a pivotal role** in effective all-in-one image restoration. Traditional approaches, often reliant on CLIP-based embeddings [43], frequently lack the robustness required for complex scenarios involving compound degradations or out-of-distribution (OOD) artifacts. To address these limitations, we leverage a VLM, such as Qwen-2.5 VL-7B [44], to perceive and reason about the degradation type, extracting explicit semantic and physical descriptors. Furthermore, as diffusion models are prone to *semantic drift* or structural inconsistency when processing severely corrupted images, the VLM is tasked with generating geospatial context tokens (τ_g). These tokens describe the underlying land-cover features and environmental characteristics, ensuring that the subsequent generative restoration remains semantically **aligned** and structurally coherent.

To facilitate informed restoration, we first perform supervised fine-tuning (SFT) on our curated *UniRS-60k* dataset. Each training sample is defined by the tuple $(\mathbf{x}, q, \tau_d, \tau_g)$, where $\mathbf{x}$ denotes the input image, q is the task-specific query, and $\{\tau_d, \tau_g\}$ represent the ground-truth degradation and geospatial context tokens, respectively. In this stage, the model is trained to interpret the imaging environment and identify degradation characteristics as a structured response.

1) Training Objective: The SFT phase optimizes the model parameters θ to maximize the likelihood of generating the joint prior tokens. Given the input pair $(\mathbf{x}, q)$, the loss function is formulated as

$$\mathcal{L}_{\mathrm{SFT}} = -\mathbb{E}_{(\mathbf{x},q,\tau_d,\tau_g)\sim\mathcal{D}} \sum_{t=1}^{T} \log \pi_\theta \left(y_t \mid \mathbf{x}, q, y_{<t}\right) \tag{1}$$

where $\mathcal{D}$ denotes the dataset, y represents the target sequence formed by the concatenation of $\{\tau_d, \tau_g\}$, and T is the sequence length.

2) Prior Inference: During the inference phase, the VLM serves as a specialized reasoning engine to predict environmental and degradation priors. Given a degraded image $\mathbf{x}$ and a prompt q, the model generates the optimal prior tokens

$$\left\{\hat{\tau}_d, \hat{\tau}_g\right\} = \arg\max_{\tau_d,\tau_g} \prod_{t=1}^{N} \pi_\theta \left(y_t \mid \mathbf{x}, q, y_{<t}\right). \tag{2}$$

The predicted degradation prior $\hat{\tau}_d$ and geospatial context $\hat{\tau}_g$ are utilized to activate the hierarchical LoRA experts in Stages II and III. Specifically, $\hat{\tau}_d$ provides the discrete routing signal required to trigger the indicator functions $\mathbb{I}(\cdot)$ in the subsequent physical-chain decoupling process. This ensures that the low-rank adapters are conditioned on precise semantic and physical interpretations, effectively mitigating the ambiguity inherent in blind image restoration.

B. Mixture-of-Experts Low-Rank Adaptation

To mitigate the prohibitive computational and memory overhead associated with full-parameter fine-tuning of large-scale models, we employ the LoRA methodology [45]. LoRA operates on the principle that weight updates during task-specific adaptation reside within a low intrinsic dimension. Formally, for a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, the adaptation is modeled by constraining the update ΔW through a low-rank decomposition. The forward pass for an input x is reformulated as

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx \tag{3}$$

where $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ are trainable matrices with rank $r \ll \min(d, k)$. During the optimization phase, W_0 remains frozen, significantly reducing the number of gradients to be stored. The matrix A is initialized using a random Gaussian distribution, while B is initialized to zero, ensuring that the initial state of the bypass branch does not alter the original model's output at the onset of training. The scaling factor α/r is applied to balance the contribution of the LoRA.

While single-adapter LoRA is efficient, it often struggles with **representation interference** in all-in-one restoration, as heterogeneous degradations (e.g., haze versus stripe noise) require conflicting weight updates. To resolve this, we propose MoLoRA, which replaces the static adapter with a set of task-specific experts governed by a routing mechanism.

As illustrated in Fig. 3, we utilize the degradation token τ_d extracted from the VLM to drive a **hard-routing strategy**. Unlike conventional MoE approaches that use soft-attention for expert blending, our routing function $\mathcal{G}(\cdot)$ performs a discrete selection to activate only the most relevant expert for a given instance, thereby minimizing intertask crosstalk. The modified forward pass for the ith layer is defined as

$$\mathbf{o}_i = W_i \mathbf{x} + \sum_{k=1}^{N} \mathbb{1}\left(\mathcal{G}\left(\tau_d\right) = k\right) \cdot \left(\frac{\alpha}{r} B_{i,k} A_{i,k} \mathbf{x}\right) \tag{4}$$

where $\mathbb{1}(\cdot)$ is the indicator function that ensures only the kth specialized LoRA expert $(B_{i,k}, A_{i,k})$ contributes to the residual branch. This design effectively decouples the parameter space across different degradation chains, allowing the framework to

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TABLE I
MAPPING BETWEEN PHYSICAL-CHAIN STAGES AND DEGRADATION TYPES

Physical Chain Stage	Underlying Physical Homology	Assigned Degradations
Atmospheric Scattering ($\mathcal{F}_{atm}$)	Reversing scattering/transmission models to restore surface reflectance	Cloud, Fog (Haze)
Sensor Acquisition ($\mathcal{F}_{sen}$)	Restoring spatial-spectral continuity and capturing low-rank structural priors	Stripe Noise, Inpainting (Missing data)
Digital Transmission ($\mathcal{F}_{trans}$)	Mitigating high-frequency loss and restoring digital signal fidelity	Gaussian Noise, Blur, JPEG Artifacts

achieve expert-level precision for diverse tasks within a unified backbone.

C. Three-Stage Training Strategy

Unified multitask training often encounters significant information interference, as heterogeneous degradation types tend to converge toward divergent optimization optima. For instance, pixel-level artifacts (e.g., Gaussian and stripe noise) necessitate fine-grained structural reconstruction, whereas semantic-level degradations (e.g., inpainting and patchwise corruption) require high-level feature synthesis. Although explicit task-conditioning via diverse guidance prompts provides a coarse-grained distinction between tasks, such a high-level semantic modulation is often insufficient to address the inherent representational conflicts arising from competing restoration objectives. However, direct task-specific adaptation via independent LoRA modules often neglects potential cross-task synergies. In data-scarce scenarios, certain degradations share inherent physical mechanisms. For instance, cloud and haze removal can be synergistically optimized by capturing shared atmospheric scattering priors. To formalize this intuition, we introduce the **Physical-Chain Prior** ($\mathcal{P}$), which provides a structured decomposition of the RS image restoration process based on the sequential physics of the imaging pipeline. Specifically, a degraded RS image I_{deg} can be modeled as the result of a nested functional chain applied to the ground truth I_{gt}

$$I_{deg} = \mathcal{F}_{trans} \circ \mathcal{F}_{sen} \circ \mathcal{F}_{atm}\left(I_{gt}\right) \tag{5}$$

where $\mathcal{F}_{atm}$, $\mathcal{F}_{sen}$, and $\mathcal{F}_{trans}$ denote the degradation functions associated with atmospheric scattering, sensor acquisition, and digital transmission, respectively. The detailed mapping between these stages and specific degradation types is summarized in Table I.

The physical-chain prior $\mathcal{P}$ defines a partition of the total task space $\mathcal{T}$ into $N_{gp} = 3$ disjoint sets: $\mathcal{T} = \{\mathcal{T}_{atm}, \mathcal{T}_{sen}, \mathcal{T}_{trans}\}$. This categorization ensures that degradations within the same group share physical homology, as they originate from the same physical layer of the imaging pipeline. It is noteworthy that our approach fundamentally deviates from conventional task grouping in two key aspects.

1) *Mechanism Versus Appearance:* Conventional grouping often clusters tasks based on visual similarity (e.g., treating "Stripe Noise" and "Gaussian Noise" collectively as "noise"). In contrast, the physical-chain prior recognizes that stripe noise is a sensor-level hardware defect requiring low-rank structural restoration, whereas Gaussian noise is a transmission-level stochastic signal loss. By decoupling them, we mitigate optimization interference between physically distinct processes.
2) *Homology Extraction:* Unlike purely data-driven grouping strategies that are susceptible to specific dataset biases, the physical-chain prior leverages inherent physical constraints (e.g., the shared radiative transfer mechanisms between cloud and haze). By aligning the model with these invariant physical laws, our approach significantly enhances generalization across diverse and unseen RS environments.

Building upon these physical insights, we propose the **HPCD** framework. This strategy progressively decomposes the restoration process across three stages—advancing from global geospatial layout learning to groupwise physical homology extraction, and finally to task-specific atomic expert refinement. This hierarchy allows the model to effectively exploit intertask correlations while simultaneously mitigating optimization conflicts, as illustrated in Fig. 2.

1) Stage I: Unified Geospatial-Semantic Prior Acquisition: To enrich the model with robust geospatial awareness and enable the discernment of Earth-surface feature variations, we perform a unified training phase across the comprehensive multidegradation dataset. This stage facilitates the injection of broad geospatial-semantic priors into the latent space of the diffusion model. Inspired by AutoDir [43], during training, we fine-tune the U-Net backbone ϵ_θ of the latent diffusion model (LDM) for all-in-one restoration. Let $\mathbf{z}_t$ denote the latent noise at timestep t, and $\mathbf{z}_I$ represent the latent embedding of the degraded image produced by the encoder $\mathcal{E}$. We employ a concatenation-based conditioning strategy to integrate the degraded image features, denoted as $[\mathbf{z}_t, \mathbf{z}_I]$. The denoising objective is defined as

$$\mathcal{L}_{LD} = \mathbb{E}_{\mathcal{E}(\mathbf{x}),\mathbf{z}_t,\mathbf{c},\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta\left(\mathbf{c}, [\mathbf{z}_t, \mathbf{z}_I], t\right)\|_2^2\right] \tag{6}$$

where $\mathbf{c}$ represents the semantic embedding derived from the VLM.

While the LDM provides a powerful generative foundation, the stochastic denoising process may compromise fine-grained structural fidelity and land-cover textures. To preserve the structural integrity of the original image, we introduce a **structure-correction module (SCM)**, inspired by [43], and denoted as $\mathcal{F}$. The SCM extracts contextual structural information from the input I to refine the intermediate prediction I_{sd} in a residual manner. To avoid the prohibitive computational cost of a full reverse denoising process, we utilize the one-step latent estimation $\tilde{\mathbf{z}}$ [46]

$$\tilde{\mathbf{z}} = \frac{\mathbf{z}_t}{\sqrt{\bar{\alpha}_t}} - \frac{\sqrt{1-\bar{\alpha}_t}\epsilon_\theta\left(\mathbf{c}, [\mathbf{z}_t, \mathbf{z}_I], t\right)}{\sqrt{\bar{\alpha}_t}} \tag{7}$$

where $\bar{\alpha}_t$ follows the noise schedule. The intermediate restored image is decoded as $I_{sd} = \mathcal{D}(\tilde{\mathbf{z}})$. The final refined output I_{res} is obtained via

$$I_{res} = I_{sd} + w \cdot \mathcal{F}\left([I_{sd}, I]\right). \tag{8}$$

The optimization of the SCM is guided by the structural fidelity loss $\mathcal{L}_{AIR}$

$$\mathcal{L}_{AIR} = \left\|I_{gt} - \left(I_{sd} + \mathcal{F}\left([I_{sd}, I]\right)\right)\right\|_2^2. \tag{9}$$

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TABLE II

ABLATION STUDY ON THE HIERARCHICAL PROGRESSIVE TRAINING STRATEGY. OUR HPCD FRAMEWORK, EVOLVING THROUGH **UNIFY PRETRAIN** (STAGE I), **GROUP TUNING** (STAGE II), AND **EXPERT REFINEMENT** (STAGE III), CONSISTENTLY IMPROVES RESTORATION FIDELITY. THIS STRUCTURED PARADIGM ACHIEVES SUPERIOR TASK COMPATIBILITY AND PERFORMANCE COMPARED TO THE **DIRECT LORA** BASELINE, PARTICULARLY IN COMPLEX ATMOSPHERIC AND SENSOR-RELATED DEGRADATIONS

Task	Direct LoRA	Unify Pretrain	Group Tuning	Expert Refine
JPEG	27.52 / 0.7492	26.79 / 0.7174	27.44 / 0.7409	**27.45 / 0.7418**
Blur	28.31 / 0.7217	27.15 / 0.6711	28.29 / 0.7364	**28.81 / 0.7526**
Inpainting	26.20 / 0.8044	28.06 / 0.7731	31.81 / 0.8481	**32.02 / 0.8594**
Gaussian	30.50 / 0.7996	31.08 / 0.8042	31.77 / 0.8342	**31.79 / 0.8350**
Stripe	33.08 / 0.8958	32.88 / 0.8949	34.97 / 0.9167	**35.62 / 0.9269**
Cloud	20.69 / 0.7913	18.58 / 0.7928	20.49 / 0.7941	**20.90 / 0.7940**
Fog	19.72 / 0.8577	20.72 / 0.8787	25.52 / 0.9164	**27.32 / 0.9213**
Avg. PSNR/SSIM	26.57 / 0.8028	26.47 / 0.7903	28.61 / 0.8266	**29.12 / 0.8333**

The total joint objective for Stage I is formulated as

$$\mathcal{L}_{total} = \mathcal{L}_{LD} + \lambda \mathcal{L}_{AIR} \tag{10}$$

where λ is a hyperparameter balancing generative performance and structural consistency.

2) Stage II: Physics-Driven Imaging-Chain Grouped Tuning (Cross-Task Synergy): To mitigate optimization conflicts between low-level pixel reconstruction and high-level semantic understanding, we propose a physics-driven grouping strategy. Rather than treating all degradations as independent entities, we leverage the *physical-chain prior* to decouple the restoration process into $N_{gp} = 3$ distinct groups: *atmospheric scattering*, *sensor acquisition*, and *digital transmission*. This stage aims to capture the shared physical homology within each group, thereby enhancing the model's robustness to diverse yet physically related degradations.

In this stage, we optimize the group-specific LoRA modules while keeping the backbone frozen. Let θ denote the pretrained parameters, and $\Delta\theta$ represents the parameter increments introduced by the grouped adapters. The training objective is formulated as

$$\mathcal{L}_{\mathrm{LD}} = \mathbb{E}_{\mathbf{z},\epsilon\sim\mathcal{N}(0,1),t}\left[\left\|\epsilon - \epsilon_{\theta+\Delta\theta}\left([\mathbf{z}_t, \mathbf{z}_I], t, \mathbf{c}\right)\right\|_2^2\right] \tag{11}$$

where $\mathbf{c}$ integrates the geospatial-textual conditions, and $\mathbf{c} = [\tau_d, \tau_g]$. To facilitate precise task adaptation, we implement a *coarse-grained routing* mechanism. Based on the degradation-specific prior τ extracted by the VLM, the adapted hidden representation $\mathbf{h}_{k+1}$ in each layer is computed by

$$\mathbf{h}_{k+1} = \sigma\left(W_k\mathbf{h}_k + \sum_{g=1}^{N_{gp}} \mathcal{G}_g\left(\tau_d\right) \cdot \left(\frac{\alpha}{r} B_k^{(g)} A_k^{(g)} \mathbf{h}_k\right)\right) \tag{12}$$

where $\mathcal{G}_g(\tau)$ is a gating function that identifies the primary physical category. In practice, $\mathcal{G}(\tau)$ acts as an indicator that selects the specific group expert matching the physical-chain prior, effectively preventing cross-category gradient interference during the joint optimization of related tasks.

3) Stage III: Task-Specific Atomic Expert Specialization (Intertask Compatibility): While Stage II effectively captures coarse-grained physical consistency, subtle gradient interference may persist between heterogeneous tasks within the same physical group (e.g., haze versus cloud). To achieve granular restoration precision, we implement a **progressive specialization** strategy, further refining the N_{gp} physical clusters into $N_{at} = 7$ atomic expert LoRA modules.

To preserve the learned physical consistency while eliminating intertask conflict, each atomic expert is initialized with the weights of its corresponding parent group adapter from Stage II. During this stage, the backbone is frozen, while only the specific atomic branch matching the identified degradation is optimized. This hierarchical transition ensures that the model exploits cross-task synergies in the earlier stage while achieving absolute task decoupling in the final stage. Based on the fine-grained task prior τ_d identified by the VLM, the adapted hidden representation is reformulated as

$$\mathbf{h}_{k+1} = \sigma\left(W_k\mathbf{h}_k + \sum_{j=1}^{N_{at}} \mathbb{I}\left(j = \mathrm{idx}\left(\tau_d\right)\right) \cdot \left(\frac{\alpha}{r} B_k^{(j)} A_k^{(j)} \mathbf{h}_k\right)\right) \tag{13}$$

where $\mathrm{idx}(\tau_d)$ denotes the task index mapped from the atomic prior τ_d, and $\mathbb{I}(\cdot)$ is an indicator function that activates the corresponding expert. This sparse activation mechanism effectively resolves optimization conflicts between competing restoration objectives by restricting the parameter update to a task-specific manifold. As demonstrated in Table II, this progressive refinement significantly outperforms direct independent LoRA training by maintaining an optimal balance between physical homology and taskwise precision.

D. Accelerated Inference via Multi-LoDMD

While diffusion models deliver high-fidelity restoration, their iterative sampling process ($T = 100$) is computationally prohibitive for real-time RS applications. To achieve industrial-grade inference speed, we propose **Multi-LoDMD** [47], which compresses the multistep denoising trajectory into a specialized **one-step generator** G_θ.

To preserve the expert knowledge acquired through the hierarchical training strategy while ensuring parameter efficiency, we freeze the pretrained U-Net backbone and exclusively distill knowledge into the **Multi-LoRA** modules. In this setting, the generator G_θ inherits the atomic expert structure from Stage III. For a specific restoration task identified by the task index $j = \mathrm{idx}(\tau_d)$, the gradient for updating the corresponding atomic LoRA parameters $\Delta\theta^{(j)}$ is defined as

$$\nabla_{\Delta\theta^{(j)}}\mathcal{L}_{\mathrm{distill}} \approx \mathbb{E}_{z,t}\Big[w_t\alpha_t\left(\epsilon_{\mathrm{fake}}\left(\mathbf{z}_t, t, \mathbf{c}\right) - \epsilon_{\mathrm{real}}\left(\mathbf{z}_t, t, \mathbf{c}\right)\right)\frac{\partial G_{\theta+\Delta\theta^{(j)}}\left(z, \mathbf{c}\right)}{\partial\Delta\theta^{(j)}}\Big] \tag{14}$$

where ϵ_{real} and ϵ_{fake} represent the noise predictions of the real and generated distributions, respectively, both conditioned on the geospatial-semantic vector $\mathbf{c}$. This formulation ensures that the distillation signal precisely optimizes the active expert module while other task-specific manifolds remain unaffected. Following the dual-model estimation strategy, we consider the following.

1) *Fixed Real Score:* ϵ_{real} is estimated via the frozen Multi-LoRA model from Stage III, providing a stable anchor that preserves the multitask expertise and generative priors.

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TABLE III

QUANTITATIVE COMPARISON ON SEVEN IMAGE RESTORATION TASKS. WE EVALUATE PERFORMANCE USING PSNR, SSIM, LPIPS, FID, AND DISTS METRICS. THE TASKS ARE ORGANIZED IN A 3–3–2 LAYOUT. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	JPEG					Blur					Inpainting				
	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓
AutoDIR ECCV 2024 [43]	26.1164	0.7044	0.3993	**1.3958**	0.3044	27.0603	0.6739	0.3500	7.0040	0.2196	21.4938	0.6264	0.5536	22.5776	0.2921
PromptIR Nips 2023 [48]	25.0852	0.7018	0.4007	**1.5621**	0.3025	22.5762	0.6256	0.4938	15.3593	0.3037	21.2822	0.6201	0.6070	24.8351	0.2967
Gridformer IJCV 2024 [49]	26.5613	0.6927	0.5465	5.9960	0.2922	25.3217	0.6355	0.5860	20.4732	0.3452	22.6377	0.6611	0.5776	4.3161	0.3273
MIR arxiv 2025 [50]	23.6392	0.6880	0.4037	1.5778	0.3029	22.1059	0.6099	0.4826	14.5977	0.2984	21.1778	0.6088	0.5963	22.6013	0.2959
HPCD-1 step (Ours)	**27.4767**	**0.7338**	**0.3694**	4.0500	**0.2345**	**27.7740**	**0.7010**	**0.3841**	**6.4480**	**0.2437**	**31.7651**	**0.8500**	**0.5342**	**0.5370**	**0.2703**
HPCD (Ours)	**27.4500**	**0.7418**	**0.3485**	2.1052	**0.2286**	**28.8098**	**0.7526**	**0.2996**	**1.6546**	**0.1937**	**32.0193**	**0.8594**	**0.5295**	**0.5325**	**0.2050**

Method	Gaussian					Stripe					Cloud				
	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓
AutoDIR ECCV 2024 [43]	32.4986	0.8295	**0.1805**	**0.4225**	**0.1426**	30.6603	0.8197	0.2439	3.0190	0.2153	18.7676	0.8277	**0.1809**	5.3925	**0.2005**
PromptIR Nips 2023 [48]	**34.0307**	**0.8740**	0.1930	2.5078	0.1647	29.5944	0.7981	0.2621	3.3147	0.2327	**21.8064**	**0.9003**	**0.0866**	2.6615	**0.0897**
Gridformer IJCV 2024 [49]	27.8722	0.6930	0.5838	7.9029	0.2755	28.2790	0.7493	0.4513	6.7393	0.2516	19.9292	0.6621	0.5895	19.0721	0.3499
MIR arxiv 2025 [50]	**33.5224**	**0.8659**	0.1878	1.9422	0.1568	27.9946	0.8111	0.2285	2.3515	0.2116	18.9894	**0.8452**	0.1078	**2.2870**	0.1010
HPCD-1 step (Ours)	31.7858	0.8350	0.1882	**0.2237**	0.1537	**35.3619**	**0.9251**	**0.0836**	**0.2044**	**0.0851**	19.0618	0.6303	0.5005	4.7028	0.3399
HPCD (Ours)	31.7858	0.8350	0.1882	**0.2237**	**0.1237**	**35.6186**	**0.9269**	**0.0775**	**0.1597**	**0.0777**	**20.8098**	0.7980	0.2441	**2.4408**	0.2118

Method	Fog (Combined)					Average					
	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	Time(s)
AutoDIR ECCV 2024 [43]	15.2230	0.7795	0.1885	7.7844	0.1615	24.5457	0.7516	0.3060	6.7995	0.2194	12
PromptIR Nips 2023 [48]	13.2489	0.7282	0.2234	11.8898	0.1986	23.9463	0.7501	0.2766	8.8758	0.2269	**0.68**
Gridformer IJCV 2024 [49]	10.7903	0.2561	0.9214	69.4068	0.3854	23.0587	0.6214	0.6080	19.1309	0.3181	1.03
MIR arxiv 2025 [50]	12.9509	0.7096	0.2399	13.4122	0.2042	22.9115	0.7341	0.2751	8.3957	0.2247	**0.62**
HPCD-1 step (Ours)	**25.6600**	**0.9109**	**0.0993**	**1.4385**	**0.0877**	**28.4113**	**0.7980**	**0.2709**	**2.5147**	**0.2021**	1.06 (0.36†)
HPCD (Ours)	**27.3188**	**0.9213**	**0.0784**	**0.3531**	**0.0646**	**29.1159**	**0.8335**	**0.2097**	**1.0673**	**0.1578**	13.2

†The proposed approach achieves a 1.06s latency with one-step diffusion generation. If the VLM is excluded, the latency decreases to 0.36s, in this scenario, our method still attains an average PSNR of 25.57, outperforming state-of-the-art models even without semantic guidance.

TABLE VI

QUANTITATIVE COMPARISON ON THE MDSD DATASET ACROSS FIVE IMAGE RESTORATION TASKS. WE EVALUATE PERFORMANCE USING PSNR, SSIM, LPIPS, FID, AND DISTS METRICS. THE TASKS AND THEIR CORRESPONDING AVERAGES ARE ORGANIZED IN A 3–3 LAYOUT. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	JPEG					Blur					Inpainting				
	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓
MIR [50]	11.6279	0.3012	0.5619	11.3656	0.3521	11.4635	0.2418	0.7466	32.8566	0.4309	18.5544	0.5796	0.5472	14.0186	0.3503
Gridformer [49]	15.2828	0.4581	0.7318	23.9559	0.3731	14.5919	0.4361	0.7799	36.0781	0.4719	19.9610	0.6361	0.7249	21.6527	0.3991
PromptIR [48]	16.9007	0.5081	0.4872	5.9336	0.3152	18.1113	0.4971	0.6467	24.5066	0.3827	18.6839	0.5788	0.3963	5.1958	**0.2484**
AutoDIR [43]	**24.4871**	**0.6832**	**0.4155**	**2.4947**	**0.2975**	**23.2023**	**0.5963**	**0.4669**	**15.5722**	**0.2798**	**19.9743**	**0.6721**	**0.3958**	**38.4744**	0.2652
HPCD (Ours)	**24.7857**	**0.6818**	**0.4210**	**4.9833**	**0.2556**	**24.6440**	**0.6565**	**0.3901**	**8.9652**	**0.2616**	**27.8886**	**0.8403**	**0.2508**	**1.0278**	**0.1774**

Method	Gaussian					Stripe					Average				
	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	DISTS↓
MIR [50]	9.0379	0.1211	0.6404	19.9160	0.4187	14.0195	0.4042	0.4152	9.7414	0.2769	12.9406	0.3296	0.5823	17.5796	0.3658
Gridformer [49]	18.3355	0.5303	0.7485	21.4923	0.3663	16.6918	0.5367	0.6696	24.0872	0.3488	16.9726	0.5195	0.7309	25.4532	0.3918
PromptIR [48]	16.4838	0.4538	0.4672	8.0510	0.2939	22.4188	0.7233	**0.2336**	3.7108	**0.1722**	18.5197	0.5522	0.4462	9.4796	0.2825
AutoDIR [43]	**27.1822**	**0.6888**	**0.3680**	**7.9638**	**0.2289**	**26.6705**	**0.6680**	0.2699	**3.5632**	0.2164	**24.3041**	**0.6617**	**0.3832**	**13.6136**	**0.2576**
HPCD (Ours)	**27.2336**	**0.7861**	**0.2628**	**1.1953**	**0.1878**	**25.8295**	**0.7862**	**0.2161**	**2.2506**	**0.1582**	**26.0823**	**0.7502**	**0.3082**	**3.6844**	**0.2081**

TABLE V

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE FOG DATASET NAMED HAZE-1K DATASET. THE RED AND BLUE COLORS INDICATE THE BEST AND THE SECOND-BEST RESULTS, RESPECTIVELY

Method	Venue	Fog_thin			Fog_medium			Fog_thick			Fog_Combined		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
ASTA [58]	GRSL'24	26.2978	0.9246	0.0473	26.1637	0.9064	0.0652	23.1796	0.8516	0.1389	24.7952	0.8733	0.0940
EMPF-Net [59]	TGRS'23	21.3244	0.8919	0.0812	24.1802	0.8931	0.0829	21.4463	0.8524	0.1454	22.5148	0.8624	0.0973
PSMB-Net [60]	TGRS'23	**27.3374**	**0.9333**	**0.0422**	27.0159	0.9094	0.0574	**24.3324**	**0.8726**	**0.1105**	25.7096	0.8891	0.0809
Trinity-Net [61]	TGRS'23	25.1596	0.9014	0.0512	26.0094	0.9089	0.0632	22.9203	0.8491	0.1795	23.9387	0.8373	0.0742
Refusion [62]	CVPR'23	24.9612	0.8964	0.0496	26.1264	0.9132	0.0604	23.2159	0.8146	0.1352	23.8737	0.8324	0.0677
RDDM [63]	CVPR'24	25.1103	0.9036	0.0569	25.8964	**0.9123**	**0.0502**	23.8694	0.8264	0.1129	24.9379	**0.9264**	**0.0671**
RSHazeDiff [64]	TITS'24	26.4394	0.9004	0.0559	**26.1943**	0.8982	0.0594	24.0146	0.8640	0.1204	**25.1140**	0.8715	**0.0645**
HPCD (Ours)	-	**27.3299**	**0.9315**	0.0544	**30.4320**	**0.9592**	**0.0347**	**24.1944**	**0.8731**	**0.1461**	**27.3188**	**0.9213**	0.0784

2) *Dynamic Fake Score:* ϵ_{fake} is modeled by a trainable denoiser μ_{fake} that tracks the evolving distribution of G_θ. It is updated concurrently using the distribution-tracking loss

$$\mathcal{L}_{\text{fake}} = \mathbb{E}_{\mathbf{z}\sim p_{\text{fake}},\epsilon,t} \left\| \mu_{\text{fake}}(\mathbf{z}_t, t, \mathbf{c}) - \epsilon \right\|_2^2 . \quad (15)$$

By calculating the discrepancy between the anchored real score and the dynamically learned fake score, the specific atomic expert within G_θ receives a distribution-matching signal that suppresses sampling artifacts. This **Multi-LoDMD** approach reconciles **cross-task synergy** with **extreme param-**

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TABLE VI

OBJECT DETECTION ROBUSTNESS ANALYSIS. OUR METHOD RESTORES DETECTION CAPABILITIES FROM COMPLETE FAILURE (ZERO VEHICLES) UNDER PATCH ATTACK TO IDENTIFYING 25 VEHICLES, COMPARED TO 58 IN THE CLEAN IMAGE

Scenario	Detected Vehicles (Count)
Clean Image (Before Attack)	58
Under Adversarial Patch Attack	0
PromptIR	0
AutoDIR	1
HPCD (Ours)	**25**
Improvement vs. Attack	**+25 Vehicles**

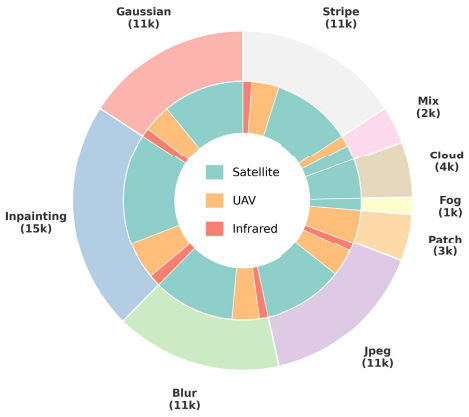


Fig. 5. Hierarchical distribution of the UniRS-60k dataset. The outer ring shows the partition of eight primary and one mixed degradation tasks, while the inner ring specifies the source platform proportions, including satellite, UAV, and infrared data.

eter efficiency, enabling high-fidelity restoration in a single forward pass and ensuring the model's readiness for large-scale industrial deployment.

E. Inference Paradigm: From Single to Composite Degradations

To accommodate the diversity of RS scenarios, we propose a bifurcated inference strategy that distinguishes between independent single degradation tasks and complex composite degradations.

1) Single Degradation Inference: When the input image is diagnosed with a specific single degradation, the framework invokes the corresponding task-specific expert from Stage III. The noise estimation $\hat{\epsilon}$ at step t is formulated as

$$\hat{\epsilon} = \epsilon_{\theta+\Delta\theta_d}\left([\mathbf{z}_t, \mathbf{z}_I], t, \tau_d, \tau_g\right) \tag{16}$$

where $\Delta\theta_d$ represents the expert LoRA weights retrieved from Stage III. The condition $\boldsymbol{\tau}_d$ is the semantic embedding derived from a specific prompt, such as: *"The most prominent distortion is strip noise."* The final restored image $\hat{x}$ is reconstructed by passing the clean latent $\mathbf{z}_0$ through the pretrained VAE decoder $\mathcal{D}$, thus $\hat{x} = \mathcal{D}(\mathbf{z}_0)$.

2) Composite Degradation Inference: In complex real-world scenarios, RS images often suffer from multiple overlapping degradations. In such cases, sequentially applying multiple Stage III experts or employing agent-based approaches may introduce interdegradation interference and cumulative artifacts. To circumvent these issues, we leverage the robust unified prior learned in Stage I, which encompasses unified geographic and physical priors derived from eight primary degradation tasks. By embedding fine-grained semantic descriptors of degradation types from the VLM into the latent diffusion space, HPCD effectively restores high-fidelity details even for complex, multilayered degradation scenarios. The noise prediction for composite cases can be defined as

$$\hat{\epsilon} = \epsilon_{\theta+\Delta\theta_{stage1}}\left([\mathbf{z}_t, \mathbf{z}_I], t, \tau_d, \tau_g\right) \tag{17}$$

where $\Delta\theta_{stage1}$ denotes the unified LoRA weights obtained during the Stage I global geographic pretraining. The condition τ_d is the descriptive diagnostic text for complex scenarios, such as: *"The most prominent distortion is strip noise and inpainting artifacts,"* and τ_g represents the geographic prior. The final image reconstruction process is the same as above. By unifying the inference paradigm with our objective function, HPCD effectively addresses both precise single-task restoration and robust composite-task recovery within a single sampling trajectory, ensuring high-fidelity reconstruction across diverse RS scenarios.

IV. EXPERIMENTS

A. UniRS-60k Dataset

Due to the scarcity of comprehensive all-in-one restoration benchmarks in the RS domain, we curated the **UniRS-60k** dataset. This large-scale collection comprises 60 000 high-resolution samples aggregated from heterogeneous RS platforms to ensure cross-domain generalization. Specifically, we integrated UAV imagery from DGA [51], satellite optical data from Haze1K [52] and RICE [53], and thermal infrared imagery from CVC-09 [54].

To simulate complex imaging environments, we incorporate a diverse array of degradations, including Gaussian noise, motion blur, stripe noise, JPEG 2000 compression artifacts, atmospheric fog, cloud cover, and localized region inpainting, alongside their complex composite forms. The simulation follows the degradation modeling strategies established by [55], [56], and [57], ensuring that the synthesized samples reflect realistic RS distortions. Furthermore, to enhance the model's robustness and its **adversarial defense capabilities**, we incorporated a specialized category termed **patch-based adversarial defense**. This task focuses on eliminating localized adversarial patches during the preprocessing stage, thereby safeguarding downstream RS image interpretation against malicious perturbations. The UniRS-60k dataset features a broad spectrum of ground sampling distances (GSDs) ranging from 0.05 to 10 m, underscoring the structural diversity and representativeness of our benchmark across multiple sensor modalities. Furthermore, the dataset encompasses an extensive variety of topographical landscapes, spanning metropolitan centers, expansive farmlands, rugged mountain ranges, and open oceans. Detailed architectural and infrastructural scenes, such as academic campuses and intricate street networks, are also incorporated to provide rich texture priors essential for high-fidelity restoration. As

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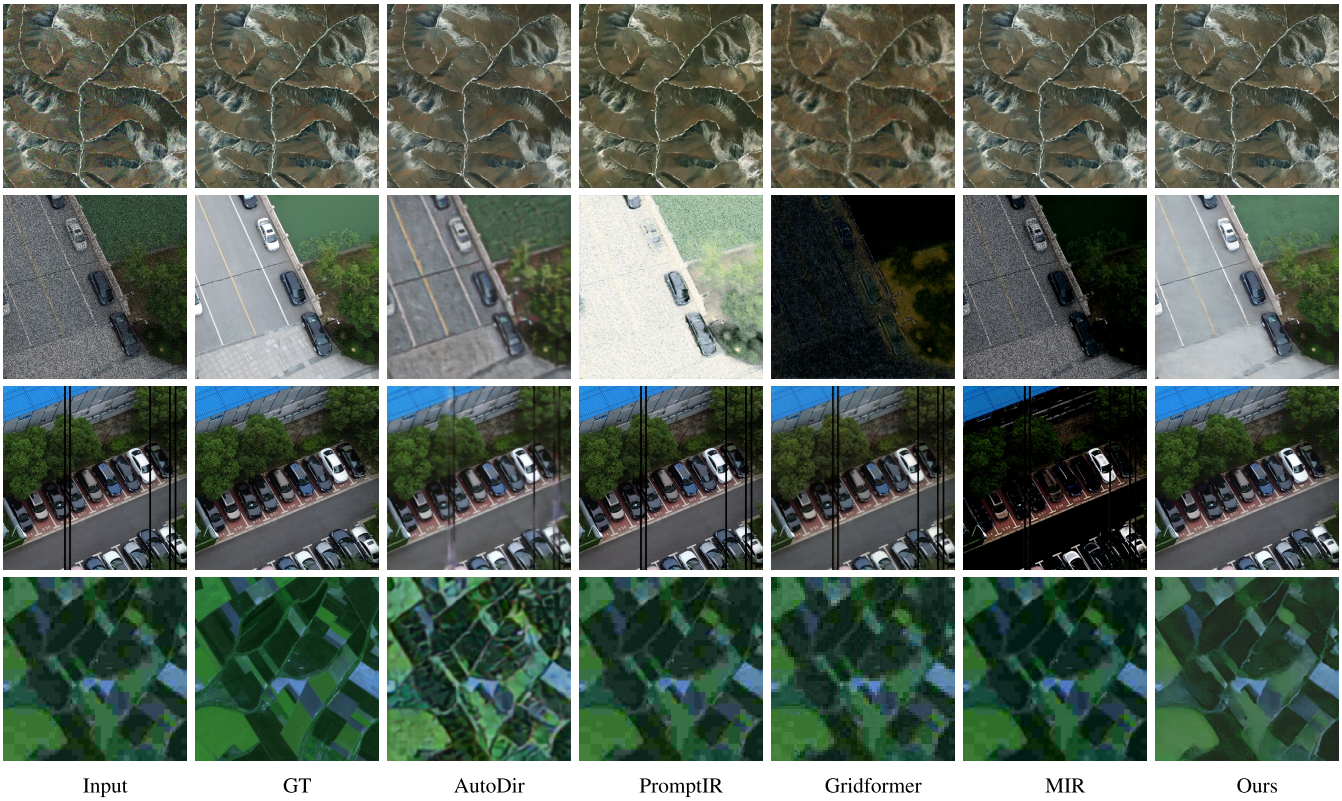


Fig. 6. Qualitative results of the HPCD model for RS-all-in-one restoration. (Top) to (Bottom) Tasks shown are: Gaussian noise removal, random inpainting, stripe inpainting, and JPEG artifacts removal.

shown in Fig 5, we provide a detailed fine-grained partitioning of the dataset's composition across each degradation category.

B. Experimental Setup and Baselines

To evaluate the effectiveness of the proposed approach, we conducted extensive experiments on three representative benchmarks, the **UniRS-60K** dataset, the **Haze-1K** dataset [52], and the **MDSD** dataset [41]. The UniRS-60K dataset was partitioned into a training set (95%) and a testing set (5%), providing a diverse range of degradation scenarios for robust model evaluation. To ensure a rigorous and fair comparison, all state-of-the-art (SOTA) baseline methods, including **AutoDIR** [43], **PromptIR** [48], **MIR-L** [50], and **Gridformer** [49], were meticulously fine-tuned on the corresponding datasets using their official implementations until full convergence. We validate our approach with several degradations, including JPEG2000, blur, inpainting, Gaussian, strip noise, cloud, and fog using our third-stage LoRA model. In addition, we validate HPCD for mix degradation and adversarial patch defense using the first-stage model. To comprehensively evaluate the effectiveness of the proposed **HPCD** framework, we conduct extensive validation across seven distinct degradation types, specifically JPEG compression artifacts, motion blur, image inpainting, Gaussian noise, stripe noise, cloud cover, and atmospheric fog. These specific tasks are addressed utilizing our third-stage low-rank adaptation experts, which are refined for atomic task precision. Furthermore, we extend the evaluation to assess the versatility of the **HPCD** framework in handling mixed degradation scenarios and adversarial patch defense. For these complex generalization tasks, we employ the first-stage model, leveraging its robust global geographic representations to maintain restoration integrity under multifaceted or antagonistic conditions.

All experiments were executed on a workstation equipped with four NVIDIA RTX 3090 GPUs to maintain consistent computational conditions across all trials. For training optimization, we employed a batch size of 6 and leveraged *Bfloat16* (BF16) mixed-precision training. This configuration was strategically selected to achieve an optimal balance between high-throughput processing and the VRAM constraints of the hardware, ensuring maximum memory efficiency without compromising numerical stability during the extensive training cycles.

C. Comparison With SOTA Approaches

1) Quantitative Comparison With Baselines: To demonstrate the efficiency and superiority of our proposed approach, we conduct extensive experiments on multiple public datasets and compare our results with several SOTA methods. As presented in Table III, we evaluate our model against leading all-in-one restoration baselines, including PromptIR [48], AutoDIR [43], Gridformer [49], and MIR [50]. These methods represent the current frontier in handling diverse degradations within a single architecture.

As evident from Table III, our proposed **HPCD** achieves the best restoration results across almost all evaluated metrics, significantly outperforming the comparative baselines.

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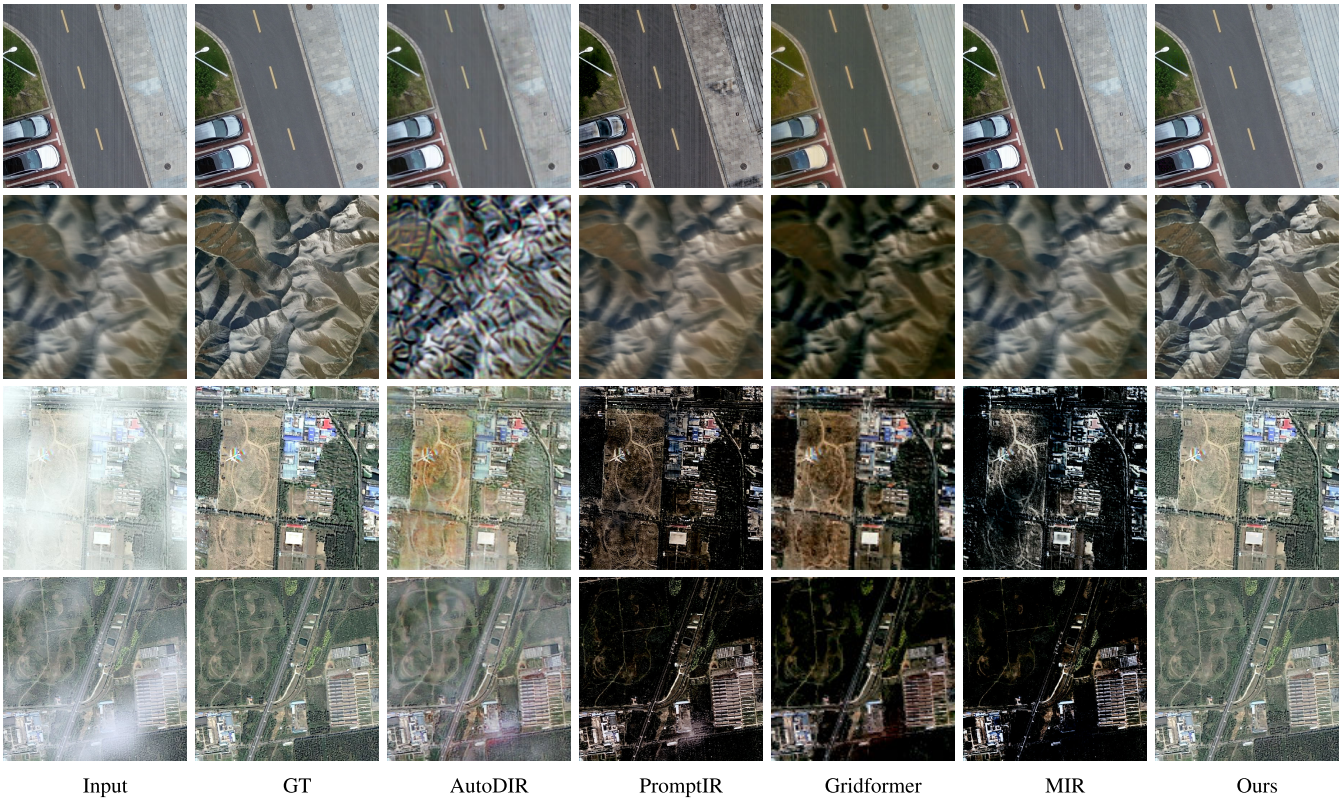


Fig. 7. Qualitative results of the HPCD model for RS-all-in-one restoration. (Top) to (Bottom) Tasks shown are: strip noise, blur, fog, and fog.

Specifically, in terms of average performance across seven tasks, HPCD attains substantial gains. Compared to the latest AutoDIR, our approach achieves an average improvement of **4.57 dB** in PSNR and **8.19%** in SSIM. Furthermore, in perception-oriented metrics, HPCD reduces the LPIPS and FID by **31.5%** and **84.3%**, respectively, which underscores its superior ability to reconstruct high-fidelity textures and maintain structural consistency.

Regarding computational efficiency, our **One-Step** variant demonstrates a remarkable speed advantage when the VLM inference time is excluded. While achieving competitive restoration quality that surpasses all baselines, it completes the inference process in a mere **0.36 s**. This represents a significant 33× **speedup** in comparison to AutoDIR (12.0 s) and even outperforms lightweight models, such as PromptIR (0.68 s), by approximately **47%**. Even when incorporating the Qwen-VL language model, our framework maintains a practical inference latency of **1.06 s**. Despite this additional overhead, it achieves superior performance relative to current SOTA methods, striking an optimal balance between high-fidelity restoration and the real-time processing requirements of RS applications.

Table IV illustrates the all-in-one restoration performance on five distinct image restoration tasks using the MDSD dataset. Table V summarizes the quantitative results on the Fog dataset (Haze-1k). Our method consistently outperforms specialized dehazing models, such as Refusion [62], RDDM [63], and RSHazeDiff [64]. Notably, in the *Fog_medium* scenario, our approach achieves a PSNR of **30.4320 dB**, surpassing the second-best method RSHazeDiff by **4.24 dB** (a 16.2% improvement). These findings underscore the importance of our HPCD in enhancing the accuracy of RS image restoration under complex environmental conditions.

2) Visual Comparison: In terms of visual quality, Figs. 6 and 7 provide a qualitative comparison of the proposed approach against other SOTA methods across various challenging restoration scenarios. As observed in Fig. 6, which showcases restoration results for **Gaussian noise, JPEG compression, and inpainting**, most compared methods struggle to effectively suppress noise while preserving fine-grained textures. For instance, AutoDIR [43] and PromptIR [48] often produce oversmoothed results in Gaussian denoising or exhibit residual blocky artifacts in JPEG restoration. In the inpainting task, these methods fail to maintain structural continuity in the missing regions. In contrast, our proposed **HPCD** demonstrates a superior ability to reconstruct RS images with sharper edges and significantly improved clarity, accurately restoring the underlying land cover details even under heavy degradation.

Fig. 7 further demonstrates the restoration performance under fog, stripe, and blur. While diffusion-based baselines, such as PromptIR and Gridformer, are prone to significant color distortion or residual haze in dense fog scenarios, our method recovers more faithful images characterized by realistic color distribution and sharper spatial edges. This superiority is primarily attributed to the integration of geographic semantic priors and our HPCD strategy, which effectively guides the generative process toward scene-consistent out-

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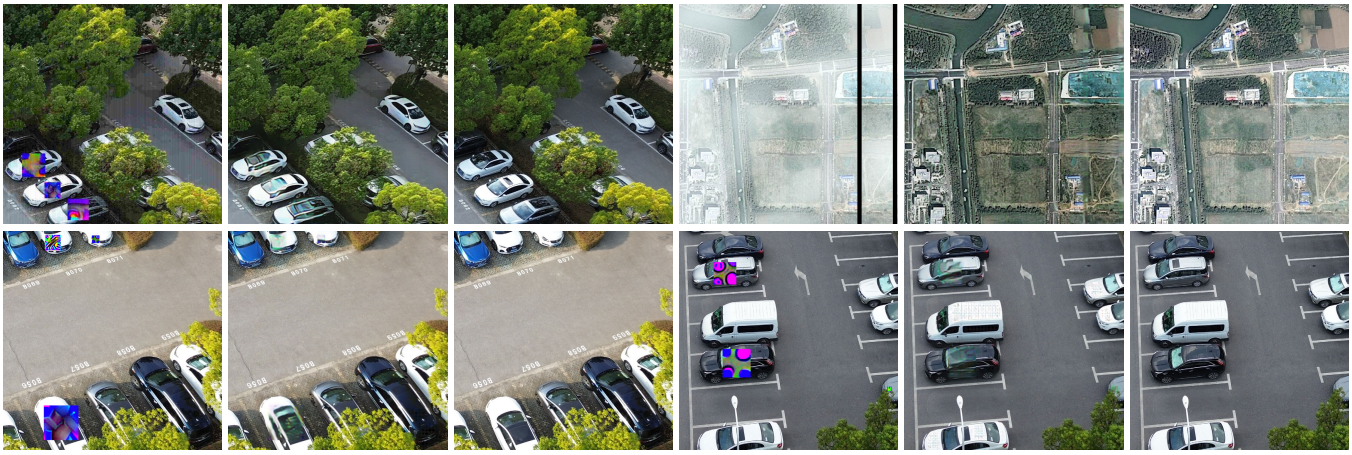


Fig. 8. Qualitative results of the HPCD model for RS-all-in-one restoration. (Top) to (Bottom) Tasks shown are: mix degradation removal (strip noise+patch-based adversarial defense+JPEG compression and fog+strip noise), and patch-based adversarial defense.

comes. Overall, our approach outperforms numerous advanced baseline methods in both quantitative and qualitative evaluations, demonstrating its robust capability in restoring high-quality RS imagery.

3) Adaptation to Mixed Degradation and Adversarial Defense: Beyond single-type restoration, we further evaluate our model's robustness under more complex conditions, including mixed degradations and adversarial attack defense.

1) *Mixed Degradation Restoration:* In real-world RS scenarios, images are often corrupted by multiple overlapping factors. We conduct experiments on several hybrid degradation combinations, such as *stripe+JPEG+attack*, *fog+strip noise*. As illustrated in Fig. 8, our proposed HPCD can effectively disentangle these complex degradations. It successfully removes nonperiodic stripes while simultaneously eliminating JPEG blocky artifacts and Gaussian noise, restoring clear and reliable land-cover information.

2) *Adversarial Patch Defense:* To explore the potential of our method in enhancing the security of downstream tasks, we investigate its performance against adversarial patch attacks. Such attacks introduce malicious patches to deceive object detectors, causing them to fail completely. As shown in Table VI, we utilize YOLOv7 [65] as the downstream detector to evaluate the robustness. In a clean scenario, the detector identifies 58 vehicles. However, under a **Patch Attack**, the detection capability drops to zero, rendering the image useless for surveillance.

Upon applying the proposed HPCD for restoration, the framework effectively mitigates adversarial patches and restores the semantic integrity of the scene. Quantitatively, the vehicle detection counts for PromptIR, AutoDIR, and HPCD after restoration are 0, 1, and 25, respectively. Notably, existing all-in-one approaches struggle to defend against localized patch perturbations, as they often fail to distinguish malicious artifacts from natural degradations. Consequently, the downstream detector is able to reidentify 25 vehicles, achieving a significant performance gain (+25) compared to the unprotected attacked images. This substantial recovery of downstream performance demonstrates that HPCD serves as a robust defense mechanism, effectively purifying adversarial inputs and safeguarding the reliability of RS analysis systems, as illustrated in Figs. 4 and 8 and Table VI.

D. Ablation Study

1) Efficacy of the HPCD Framework and Three-Stage Training Strategy: To validate the superiority of our HPCD approach, we conduct a comparative analysis against a standard Direct LoRA training baseline. As illustrated in Table II, our proposed methodology significantly outperforms the direct LoRA approach, achieving an average PSNR gain of 2.55 dB. This substantial improvement demonstrates the effectiveness of simultaneously exploiting physical homology and task heterogeneity. Unlike direct LoRA, which often suffers from task interference and suboptimal convergence in multidegradation scenarios, our HPCD approach utilizes a structured hierarchy to capture shared physical representations while preserving task-specific nuances. By aligning the decoupling process with the physical imaging chain, our method maximizes cross-task synergy, particularly in categories like atmospheric scattering and sensor artifacts, where shared priors are crucial for accurate reconstruction.

Furthermore, Table II details the performance evolution across our three-stage training recipe, showing a clear progressive improvement in image restoration quality as the average PSNR rises consistently from 26.47 dB in Stage I to 29.12 dB in the final stage. Specifically, Stage I (unify pretrain) focuses on acquiring global geographic-semantic priors, establishing a stable foundation of the geographic layout across the entire dataset. Subsequently, Stage II (group tuning) introduces physics-driven grouped tuning, where the model learns physics-grouped shared features, such as joint priors for cloud and fog. This stage yields the most significant performance leap from 26.47 to 28.61 dB, confirming that exploiting physical homology effectively reduces the learning difficulty for complex degradations. Finally, Stage III (expert refinement) executes task-specific atomic expert refinement, where experts are fine-tuned to capture granular

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TABLE VII

ROBUSTNESS ANALYSIS OF GEOGRAPHIC PRIORS UNDER EMBEDDING NOISE. COMPARED TO THE OPTIMAL CONFIGURATION, THE MODEL MAINTAINING CONSISTENT PERFORMANCE UNDER NOISE INJECTION DEMONSTRATES THE STABILITY OF OUR VLM-GUIDED REGULARIZATION

Task	Ours w/o geo. priors	Ours w. noise	Ours
JPEG	26.04 / 0.7046	26.48 / 0.7205	**27.45 / 0.7418**
Blur	28.68 / 0.7378	28.75 / 0.7434	**28.81 / 0.7526**
Inpainting	28.26 / 0.8150	28.50 / 0.7928	**32.02 / 0.8594**
Gaussian	30.28 / 0.8148	30.19 / 0.8065	**31.79 / 0.8350**
Stripe	31.95 / 0.8941	31.95 / 0.8886	**35.62 / 0.9269**
Cloud	20.39 / 0/7782	20.55 / 0.7824	**20.90 / 0.7940**
Fog	19.62 / 0.6089	24.29 / 0.8995	**27.32 / 0.9213**
Avg. PSNR/SSIM	26.46/0.7647	27.24 / 0.8048	**29.12 / 0.8331**

task details, further elevating the average PSNR to 29.12 dB and achieving expert-level precision across diverse RS tasks.

2) Robustness of Geographic Priors: To validate the contribution of geographic priors, we conducted an ablation study by excluding the geographic component τ_g during inference. The results, summarized in Table VII, demonstrate that the inclusion of geographic priors significantly enhances the quality of the restored images, with the average PSNR increasing from 26.46 to 29.12 dB. To evaluate the stability and reliability of the geographic context priors provided by our VLM-guided module, we conduct a robustness test by injecting Gaussian noise into the text embeddings generated by the text encoder. As shown in Table VII, while the introduction of noise leads to a marginal degradation in performance compared to the original HPCD model, the metrics remain remarkably stable across diverse tasks. We observe that tasks involving complex structural restoration, such as *Stripe* and *Inpainting*, exhibit higher sensitivity to embedding noise compared to atmospheric tasks like *Cloud*. This phenomenon can be attributed to the fact that these specific restoration processes rely more heavily on the precise geometric structure guidance and spatial layout constraints inferred from the VLM. Nevertheless, the overall framework maintains a high average PSNR of 27.24 dB, demonstrating that our HPCD framework leverages semantic geographic context as a robust structural constraint rather than an unstable prior. This resilience is crucial for real-world RS applications where VLM-inferred priors may contain inherent uncertainty.

E. Discussion

1) Discussion on Model Scaling and Quantization: To further reduce the computational latency and enhance the practical deployment feasibility of the VLM component, we conducted a comprehensive investigation into various model scales and quantization schemes. Specifically, we benchmarked the 3B variant (Qwen2.5-VL-3B-Instruct) against a 4-bit-quantized 7B version (Qwen2.5-VL-7B-Instruct-AWQ). To ensure a fair comparison, both models were fine-tuned using the same methodology applied to the baseline Qwen2.5-VL-7B. As summarized in Table VIII, the empirical results reveal that the AWQ-quantized 7B model strikes an optimal balance between restoration fidelity and inference efficiency. Notably, it delivers highly competitive PSNR performance while significantly reducing latency compared to its full-precision 7B counterpart, thereby justifying its selection for latency-sensitive applications.

TABLE VIII

ABLATION STUDY OF HPCD WITH DIFFERENT VLM CONFIGURATIONS

Variant	PSNR↑	SSIM↑	Time(s)
HPCD w/o VLM (1-step)	25.5732	0.7643	**0.36**
HPCD-1 step (Qwen2.5-VL-3B)	27.0132	0.7803	0.73
HPCD-1 step (Qwen2.5-VL-7B-AWQ)	27.8316	0.7908	0.88
HPCD-1 step (Qwen2.5-VL-7B)	28.4113	0.7980	1.06
HPCD	**29.1159**	**0.8335**	13.2

TABLE IX

QUANTITATIVE EVALUATION ON REAL-WORLD RESTORATION BENCHMARKS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Task	Method	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
Fog (Combined)	MIR (arXiv 2025)	12.9509	0.7096	0.2399	13.4122
	PromptIR (NeurIPS 2023)	13.2489	0.7282	0.2234	11.8898
	AutoDIR (ECCV 2024)	15.2230	0.7795	0.1885	7.7844
	HPCD (Ours)	**27.3188**	**0.9213**	**0.0784**	**0.3531**
Cloud	MIR (arXiv 2025)	18.9894	**0.8452**	**0.1078**	**2.2870**
	PromptIR (NeurIPS 2023)	**21.8064**	**0.9003**	**0.0866**	2.6615
	AutoDIR (ECCV 2024)	18.7676	0.8277	0.1809	5.3925
	HPCD (Ours)	**20.8098**	0.7980	0.2441	**2.4408**
Real Blur (GoPro)	MIR (arXiv 2025)	24.8907	0.8296	0.1795	34.3541
	PromptIR (NeurIPS 2023)	27.6358	0.8575	0.1876	39.7552
	AutoDIR (ECCV 2024)	**26.9267**	**0.8762**	**0.1079**	**28.3763**
	HPCD (Ours)	**27.0988**	**0.8966**	**0.1261**	**32.3747**
Real Noise (SIDD)	MIR (arXiv 2025)	23.0687	0.3439	0.7744	**156.7363**
	PromptIR (NeurIPS 2023)	24.9730	0.4780	0.7001	176.2245
	AutoDIR (ECCV 2024)	**25.5838**	0.4597	**0.6957**	**157.4193**
	HPCD (Ours)	**25.9204**	**0.5243**	**0.5580**	173.7013

2) Generalization to Complex Real-World Scenarios: To further validate the efficacy of our approach, we conducted supplementary experiments to bridge the synthetic-to-real gap. While five degradation types in UniRS-60k (e.g., Gaussian noise) are synthesized following established protocols [55], [56], [57], the cloud and fog categories are curated from authentic real-world captures to preserve physical consistency. To rigorously assess the model's performance on authentic sensor-level degradations, we extended our evaluation to renowned real-world benchmarks, including GoPro for motion blur [66], SIDD for sensor noise [67], and a specialized striping noise dataset [68]. Specifically, 100 samples were randomly selected from each benchmark to verify the model's robustness in practical settings. For JPEG compression and inpainting, we do not explicitly distinguish between simulated and real-world cases, as their underlying degradation mechanisms—namely, quantization and structural occlusion—remain intrinsically consistent across both domains. These extensive evaluations, illustrated in Table IX and Fig. 9, underscore the robustness of our HPCD framework in addressing intricate, practical RS degradations.

3) Compared to Multiagent Approaches: To influence the advantage of our HPCD method over agent-based approaches. We compared with agent-based approaches. As there are currently no established agent-based frameworks specifically for RS image restoration, we implemented a multiagent baseline to provide a comprehensive comparison. This framework utilizes a fine-tuned Qwen2.5-VL-7B to identify degradation types and invoke corresponding specialized tools. The toolset com-

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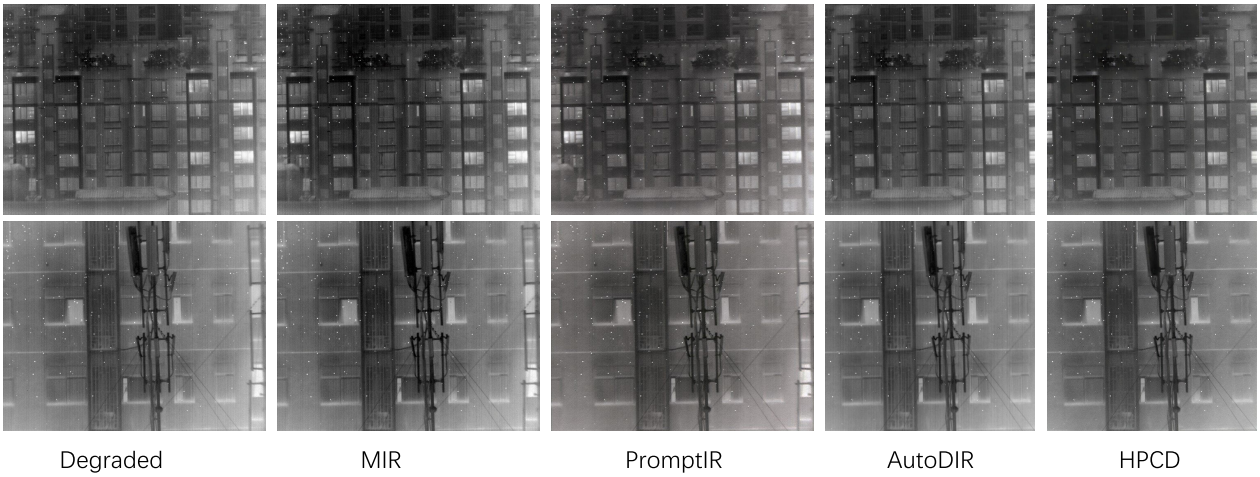


Fig. 9. Qualitative comparison of different restoration algorithms for strip noise removal in real-world scenarios. (Left to right): degraded images, results of MIR, PromptIR, AutoDIR, and our proposed HPCD model.

prises several representative RS restoration methods, including DehazeFormer [37] for dehazing, Restormer [10] for Gaussian noise and blur, FBCNN [69] for JPEG artifacts, LAMADA [70] for inpainting, ASCNet [68] for destriping, and ACANet [71] for cloud removal. As demonstrated in Table X, our proposed HPCD consistently outperforms this multiagent approach. Through practical evaluation, we observed that the performance of the agent-based system is highly sensitive to the diagnostic accuracy of the VLM. Misidentification of degradation types, **such as mistaking inpainting artifacts for stripe noise or confusing thick fog with clouds**, leads to the invocation of incorrect tools and significantly degrades average performance. In contrast, our HPCD framework exhibits superior robustness against such misidentifications. Even if confusion occurs between strip and inpainting, the restoration quality remains stable because the respective LoRA modules are derived from the same sensor-acquisition-related base through continuous training. **Furthermore, the agent-based approach struggles with thick fog due to its visual similarity to clouds and the lack of synergistic training between these tasks. Our method benefits from joint learning on fog and cloud removal, as both processes involve reversing atmospheric scattering to restore surface reflectance. We have incorporated this analysis into Section IV-E, this visual results compared to multi-agent approaches are shown as in Fig. 10.**

TABLE X

QUANTITATIVE COMPARISON OF RESTORATION PERFORMANCE BETWEEN THE AGENT-BASED APPROACH AND OUR PROPOSED HPCD ACROSS SEVEN PRIMARY DEGRADATION TASKS. BOLD VALUES INDICATE SUPERIOR PERFORMANCE

Type	Agent-based		Ours (HPCD)	
	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
JPEG	**27.69**	**0.7501**	27.45	0.7418
Blur	26.31	0.6026	**28.81**	**0.7526**
Inpainting	16.55	0.5180	**32.02**	**0.8594**
Gaussian	20.19	0.4288	**31.79**	**0.8350**
Stripe	30.43	0.7579	**35.62**	**0.9269**
Fog	23.88	0.7136	**27.32**	**0.9213**
Cloud	**22.10**	0.7853	20.90	**0.7940**
Average	23.88	0.6509	**29.13**	**0.8330**

TABLE XI

QUANTITATIVE EVALUATION ON THE **RANDOM MIXED DEGRADATION** BENCHMARK. EACH TEST IMAGE IS CORRUPTED BY 2–3 RANDOMLY SELECTED DEGRADATION TYPES (FROM SEVEN CATEGORIES). METRICS ARE AVERAGED OVER 100 SAMPLES. BOLD VALUES INDICATE SUPERIOR PERFORMANCE

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
PromptIR	13.54	0.3021	0.6391	201.47
MIR	15.05	0.4153	0.6295	200.91
AutoDIR	15.39	0.4317	0.6187	194.09
Ours (HPCD)	**18.14**	**0.4676**	**0.5578**	**175.84**

4) Quantitative Results Over Mix Degradation Benchmark: To systematically evaluate the performance of HPCD under complex scenarios, we constructed a random mixed degradation benchmark. In this setup, each test image is simultaneously corrupted by 2–3 degradation types randomly selected from seven categories, creating a vast space of unpredictable and combinatorial physical interference. As summarized in Table XI, existing SOTA methods, such as PromptIR and AutoDIR, exhibit significant performance degradation in this challenging scenario. In contrast, our HPCD maintains a substantial lead, achieving a PSNR of 18.14 dB and an FID of 175.84. These superior results indicate that our robust "global layout" reconstruction effectively prevents the model from being overwhelmed by coupled degradations.

5) Discussion on DMD Distillation Across Diverse Degradation Tasks: As demonstrated in Table III, the stripe expert achieves superior distillation performance, whereas a noticeable gap in FID persists between the one-step and 100-step configurations for fog and cloud tasks. To elucidate this phenomenon, we analyze the training dynamics of representative atomic experts, as illustrated in Fig. 11. Here, L_{KL} measures the Kullback–Leibler (KL) divergence between the

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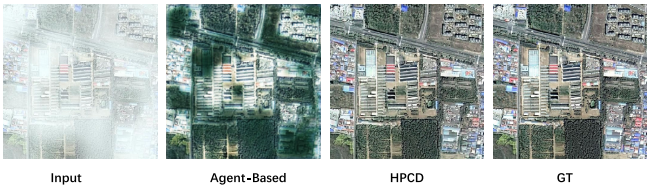


Fig. 10. Quantitative comparison between the proposed HPCD and the multiagent framework on fog removal tasks.

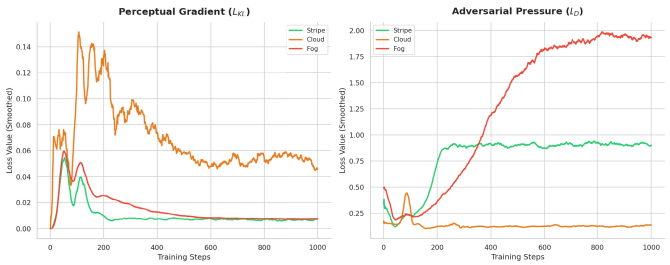


Fig. 11. Distillation dynamics and loss convergence of representative atomic experts.

distilled student and the teacher model distributions, while L_D signifies the adversarial pressure exerted by the discriminator in distinguishing generated samples from real data. **The L_{KL} for the stripe expert converges rapidly to a near-zero baseline while maintaining stable and moderate adversarial pressure (L_D).** This signifies a favorable equilibrium where the generator effectively captures the underlying distribution, reaching a professional level of realism with minimal discrepancy from the teacher. In contrast, the cloud and fog experts exhibit distinct training behaviors that account for their relative performance degradation. Specifically, the **perceptual gradient (L_{KL}) for the cloud expert displays significantly higher values and more intense oscillations compared to Stripe.** This suggests that the stochastic and nonuniform nature of cloud textures poses a substantial challenge for consistent feature alignment between the one-step student and the multistep teacher. Simultaneously, the adversarial pressure (L_D) for the fog expert escalates significantly during the late stages of distillation. This trend reveals an inherent representation bottleneck, where the generator produces challenging samples that the discriminator identifies as lacking realism, yet the one-step model struggles to synthesize the large-scale, continuous atmospheric textures required to resolve this conflict. The underlying cause for these difficulties is that cloud and fog scenarios involve complex ground details and spatially variant degradations. Such intricate physical priors make it inherently difficult for a compressed single-step denoising trajectory to achieve the rapid convergence and high-fidelity perceptual reconstruction attainable in simpler, spatially uniform tasks.

V. Conclusion

This article introduces an HPCD framework that addresses the fundamental tension between physical homology and task heterogeneity in all-in-one RS image restoration. By partitioning the restoration pipeline into three progressive stages, namely global layout learning, physics-driven group tuning, and task-specific refinement, our approach systematically aligns with the RS imaging chain to maximize cross-task synergy while mitigating negative transfer. Furthermore, the integration of VLM-guided regularization and Multi-LoDMD distillation ensures high semantic fidelity and facilitates efficient one-step inference. Comprehensive evaluations on the UniRS-60K, MDSD, and Haze-1K benchmarks demonstrate that HPCD achieves SOTA precision across seven degradation types and various cross-modal scenarios. The framework provides a robust and scalable solution for high-quality RS scene reconstruction and serves as a promising paradigm for unified restoration in complex environmental contexts.

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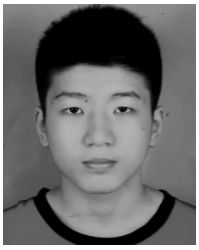
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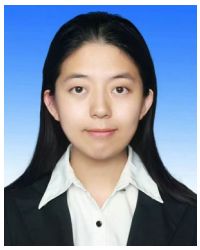
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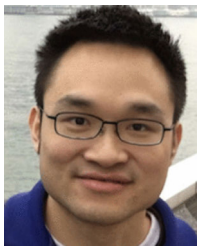
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